LEARNING TO IGNORE WHAT MATTERS: REPRESENTATION, PREDICTION, AND CONTROL IN JEPA WORLD MODELS

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ABSTRACT

A world model should discard visual detail without discarding the distinctions an agent needs to act. Joint-embedding prediction makes this trade-off implicit: it learns both the representation and the prediction target. We examine three separate questions: what information remains encoded, what survives prediction, and what supports control. We begin with Craftax, where one 512-dimensional joint-embedding model reaches 121 of 128 nearby goals in a six-action open-loop test, yet its tested reward-planning pipeline underperforms random actions. A controlled moving-wall task then separates representation from imagination: decoded real encodings retain the wall, and policies using them exceed 50% survival in two of nine runs; decoded predicted rollouts lose the wall, and none of 27 policies trained in the tested imagination pipeline reaches that threshold. These pipeline outcomes include fitted heads and policy optimization and do not isolate dynamics as the cause. Finally, nuisance-texture worlds connect the observations to the objective. We formalize a known predictable-nuisance failure under Gaussian non-collapse, and construct an exactly Gaussian, fully decodable representation in which a control-relevant bit has arbitrarily small prediction-loss scale. The resulting lesson is a concrete evaluation requirement: retention, prediction, and control must be checked separately.

1 INTRODUCTION

A useful world model need not reproduce every pixel. It must preserve the distinctions that change what an agent should do. For a robot approaching a doorway, the texture of the wall may be irrelevant; the wall's position and the available opening are not. Learning this separation without specifying a particular downstream reward is a central attraction of joint-embedding predictive architectures (JEPAs): predict an informative representation of the future rather than reconstruct the future image (LeCun, 2022; Assran et al., 2023; Bardes et al., 2024).

Recent systems make this approach increasingly practical. Pretrained video representations support action-conditioned planning, and end-to-end objectives jointly learn representations and dynamics (Assran et al., 2025; Sobal et al., 2025; Maes et al., 2026). These successes raise an evaluation question: *when a learned world model supports one behavior, what has that established about its usefulness for another?* Nearby goal reaching, recognizable decoded rollouts, and reward-driven control measure different capabilities. A representation can encode a feature that its predictor does not preserve; a predictor can achieve low error on features that a controller cannot use.

We study these distinctions in a squared-loss JEPA with Gaussian marginal regularization, following the end-to-end formulation of LeWorldModel (Maes et al., 2026; Balestriero & LeCun, 2025). The study moves from a complex environment to progressively simpler ones (Figure 1). Craftax first exposes a split between short-horizon goal reaching and reward control. A moving-wall task reduces the decision to one observable obstacle and an initially hidden gap. Cube worlds then remove the obstacle and vary only visual nuisance and dynamics. The simplification makes the question sharper: is relevant information absent from the encoding, lost during prediction, or ineffective in the complete control pipeline?

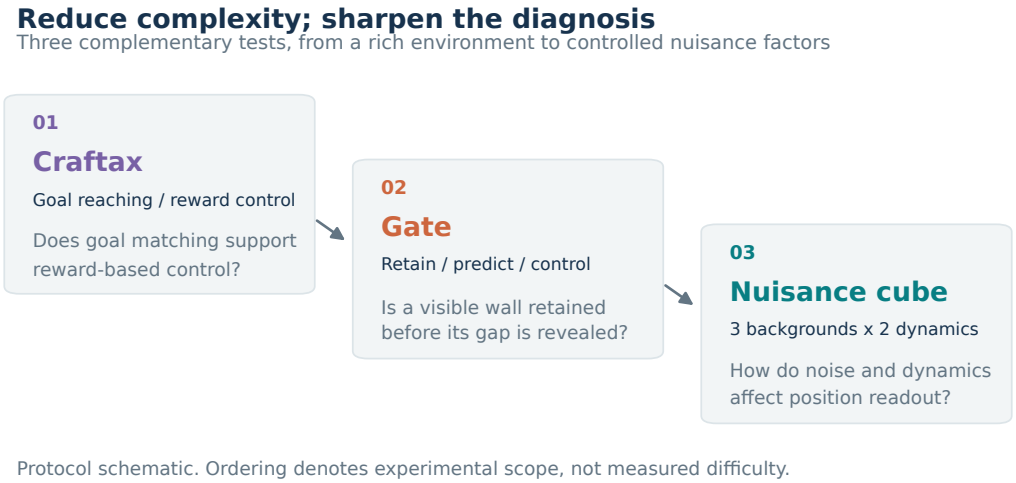


Figure 1: **Simplify the environment to sharpen the question.** Craftax tests whether local visual-goal planning transfers to reward-driven behavior. The gate exposes a single control-relevant obstacle and separates encoded observations from imagined futures. Cube worlds isolate predictable nuisance. The environments supply complementary diagnostics, not a shared performance leaderboard.

The gate provides the clearest empirical separation. A decoder recovers the wall from frozen real encodings, and a convolutional policy trained on those decoded observations discovers the survival strategy in two of nine runs. The same decoder applied to predicted latents loses the wall within two steps for all three fitted encoders. None of the 27 policies trained in the tested JEPA imagination pipeline discovers the strategy. The decoder adds supervision, and imagined training also changes reward estimation and policy optimization; these distinctions are essential to what the experiment establishes.

Our contributions are threefold. First, we organize the evidence around separate tests of **retention, prediction, and control**, making clear which transfers between these tests are supported. Second, we provide a controlled, replicated gate study showing useful information in a frozen representation alongside poor performance of its imagination pipeline, with predictor variants and model seeds reported separately. Third, we characterize two limits of the training objective: Gaussian non-collapse can coexist with nuisance-only encoding, and exact decodability can coexist with arbitrarily small prediction-loss scale. The predictable-nuisance mechanism is known (Sobal et al., 2022); our theory specializes it to Gaussian regularization and separates it from the distinct question of how much loss is assigned to retained information.

2 What is learned, and what is evaluated?

Joint representation and prediction. An encoder maps an image to $z_t = f_\theta(x_t) \in \mathbb{R}^D$. An action-conditioned predictor uses H latent frames and their actions to predict the next latent:

$$\mathcal{L}(\theta) = \frac{1}{D}\widehat{\mathbb{E}}_t \left\| g_\theta(z_{t-H+1:t}, a_{t-H+1:t}) - z_{t+1} \right\|^2 + \lambda \,\mathrm{SIGReg}(\{z_t\}). \tag{1}$$

Gradients reach both the predictions and their targets; there is no stop-gradient or teacher. SIGReg compares the characteristic functions of random latent projections with those of a standard Gaussian (Balestriero & LeCun, 2025). It constrains the marginal distribution of features, not which physical variables they represent. The precise empirical and population expressions appear in Appendix A.

The main grid model uses a $4 \times 4 \times 32$ latent ($D = 512$), a tile-transformer encoder, and a factored spatial–temporal predictor with eight-frame history. Its representation training uses neither pixels as prediction targets nor rewards. Each model is fitted on 500,000 transitions for approximately ten minutes on an A100; post-hoc decoders, heads, and policies are fitted separately. The Craftax and

Table 1: **Different tasks, one Craftax checkpoint.** Goal reaching uses 128 nearby goals and six open-loop actions. Reward control uses 128 starts and 100 steps; $\pm$ is the standard deviation across starts. These are exploratory single-model results, not the standard Craftax achievement benchmark.

Test	Controller	Result
Goals	JEPA latent-goal planner	$121/128$ reached
	Random actions	$24/128$ reached
Return	Random actions	1.45 ± 1.09
	Myopic fitted reward	0.78 ± 1.11
	One-step prediction + value	0.94 ± 1.20
	Six-step CEM + value	0.47 ± 1.10

gate checkpoints use the same grid shape and training family but different data. Appendix C gives the complete settings and measured times.

Three tests. A frozen decoder tests whether information is *recoverable from an encoding* under additional supervision. Applying that same decoder to predicted latents tests whether it remains accessible in *model-generated futures*; because such latents can leave the encoder's distribution, a bad rendering alone does not prove that the information is absent. Finally, a controller evaluated in the real environment tests the *complete learned pipeline*, including its observation interface, reward and termination estimates, optimizer, and training distribution. These tests become more informative when their shared components and changed components are explicit.

Reference model. For the gate, a reconstruction-trained tokenizer and causal flow-matching predictor provide a feasibility reference (Lipman et al., 2023; Chen et al., 2024; Hafner et al., 2025). It uses the same latent shape, data, and nominal model-training budget, with a 32-frame context and jointly fitted reward and termination heads. Architecture, supervision, and history length differ from the JEPA. One dynamics checkpoint is evaluated, so its policy runs show variation conditional on that fit rather than variation across fitted flow models.

3 Craftax: local goals and reward control ask different questions

Craftax-Classic combines movement, resources, inventory, and survival (Matthews et al., 2024). It is a useful starting point because reaching a nearby visual target and selecting a rewarding sequence of interactions can require different information. We evaluate one fitted 512-dimensional grid JEPA in two protocols, both using real simulator outcomes.

Short-horizon goal reaching. After seven random warm-up actions, six additional random actions generate a reachable goal image. The planner minimizes distance between its predicted endpoint latent and the encoded goal. It then executes the selected six actions *open loop*, without replanning. Success means matching the goal's final player position; it does not require matching inventory or the entire scene. The planner reaches 121 of 128 goals, compared with 24 for random actions (Table 1). Goal generation and execution share simulator randomness. This is evidence for nearby goal reaching under the stated protocol, not long-horizon task completion.

Reward-driven planning. We separately fit reward and value heads on frozen encodings and evaluate 100-step behavior from 128 starts. The six-step CEM controller replans after every real observation. Its mean return is 0.47, below 1.45 for uniform random actions. A myopic reward controller, which does not use predicted transitions, also underperforms random actions. The reward head's correlation falls from 0.92 on its fitting data to 0.35 on held-out time rows, and value correlation falls from 0.99 to 0.27. Poor performance therefore already appears in reward-based action selection without predicted transitions; the multi-step result concerns the combined head, dynamics, and planning pipeline.

This split motivates the controlled studies that follow. Goal-reaching success shows that a model can guide some actions; it does not certify reward prediction or policy learning in that model. Conversely, a weak reward pipeline does not establish that its encoder discarded the relevant state. Appendix H records the planner and head-fitting details.

4 THE GATE: INFORMATION IN THE ENCODER AND IN IMAGINATION

A decision with a visible cause. A red cube moves in a 63×63 image. A vertical wall approaches from the right; its gap is at the top or bottom with equal probability, but remains hidden until the wall passes a fixed reveal point. From the centre, the gap becomes visible only two steps before contact, while either gap requires five vertical moves. Retreating creates time to observe the gap and respond. Contact with a wall or border ends the episode with reward -1; all other rewards are zero. Episodes last at most 128 steps. The model-fitting data contain $500{,}000$ uniform-random transitions, with a separate $25{,}000$-transition evaluation archive.

Comparisons and replication. We fit three independent grid-JEPA encoders. For each, recurrent PPO (Schulman et al., 2017) is trained in the real gate on frozen latents, and inside the frozen world model on predicted latents. The latent policies use the same MLP torso design. In imagination, rewards and terminations come from fitted heads; in the real gate they come from the simulator. Collection length, start states, update organization, and step budgets also differ (Appendix C), making this a comparison of complete training pipelines. Its real-versus-imagined contrast is not a dynamics-only intervention.

Each original predictor has three policy seeds per encoder, giving nine policies. A fitted conditional residual predictor is evaluated with one sample and with a mean of eight samples, adding nine policies per variant. Real-latent controls have nine policies; real-pixel controls have three per torso. Every reported policy is its final checkpoint, evaluated on 256 real episodes. The primary metric is walls passed per episode. We call survival above 50% *discovery* of the survival strategy. Model fits and policy seeds are separate replication levels; the real and imagined evaluations use different random-key schedules and are not paired trials.

4.1 USEFUL WALL INFORMATION REMAINS IN THE FROZEN REPRESENTATION

Decoded real encodings visibly retain the approaching wall (Figure 2). To test whether that information can support behavior, we train a convolutional policy in the real gate on the frozen decoder's output. Two of nine runs discover the survival strategy, both using encoder seed 1; the corresponding real-pixel CNN discovers it in two of three runs (Figure 3). This establishes useful information in at least one fitted encoder under the decoder-and-CNN read-out. The decoder adds reconstruction supervision outside the world-model budget, so this is a test of recoverable information rather than unsupervised control sufficiency.

The policy interface matters. With an MLP torso, real pixels yield 0.91 walls per episode and frozen latents yield 0.94, with no discoveries in either group. An MLP on decoded latents similarly yields 0.92 walls. Thus failure of this MLP training recipe is not evidence that the encoder has lost the obstacle. These comparisons do not set a capacity limit for all MLP policies; they locate a limitation of the tested read-out and optimization procedure.

4.2 THE TESTED IMAGINATION PIPELINE DOES NOT DISCOVER THE STRATEGY

All three JEPA predictor variants produce poor downstream control (Table 2; per-seed outcomes in Appendix F). The original predictor yields 0.27 walls per episode; one residual sample and the mean of eight each yield 0.26. No policy among these 27 runs discovers the strategy; the highest final survival is 3.1%. Real-transition PPO on the same frozen encoders yields 0.94 walls. The difference measures the combined effect of learned transitions, fitted heads, and the changed PPO protocol. Even the stronger real-latent MLP result falls far short of reliable survival.

The flow reference supports feasibility at this task and nominal model budget: one of its nine policies discovers the strategy. The remaining eight pass 0.72–1.18 walls per episode. Since all nine use one dynamics fit with a different architecture and supervision, this reference does not establish a model-

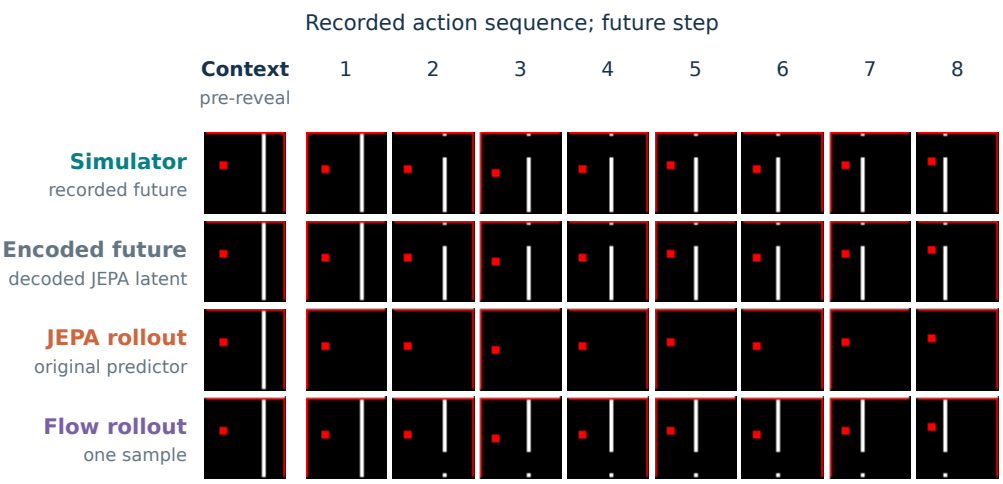


Figure 2: **A wall visible in real encodings disappears in predicted rollouts.** A pre-reveal context is followed by futures under recorded actions: the simulator, decoded real JEPA encodings, decoded original JEPA predictions, and one flow-model sample. The JEPA rows use encoder seed 0 and the same fitted decoder. The example is the first stored diagnostic context at wall position $x = 45$ among 64 contexts whose recorded futures survive the horizon. The flow sample is a single-checkpoint feasibility reference. A decoding failure on predicted latents can reflect both prediction error and off-distribution read-out; it is not alone proof of information loss.

Table 2: **Final gate outcomes, with predictor variants kept separate.** Mean $\pm$ sample standard deviation across policy runs; discovery is final survival above 50%. JEPA imagination variants and real-latent conditions each contain 3 models $\times$ 3 policies; pixel controls have three policies. Real training changes rewards, terminations, and PPO protocol alongside dynamics. Flow is a single-model reference.

Training / observation	Predictor or torso	Walls	Discovery
Imagined JEPA	Original	0.27 ± 0.22	$0/9$
	Residual sample	0.26 ± 0.32	$0/9$
	Residual mean of 8	0.26 ± 0.30	$0/9$
Real / frozen JEPA	MLP	0.94 ± 0.04	$0/9$
Real / pixels	MLP	0.91 ± 0.08	$0/3$
Real / decoded JEPA	CNN	1.57 ± 1.26	$2/9$
Real / pixels	CNN	2.91 ± 1.65	$2/3$
Imagined flow	Sample	1.45 ± 1.23	$1/6$
	Mean of 8	0.88 ± 0.14	$0/3$

family ranking. Training the gate JEPA for twice and four times the steps also leaves imagined-policy survival at 0–3% in the tested extensions (encoder seed 0, policy seed 0, each predictor variant).

4.3 Prediction and fitted heads provide distinct failure signals

The same decoder that renders real encoded walls removes them from original predicted rollouts. Across all three encoder seeds, decoded shared-wall intensity is at most 0.02 by the second predicted step (Appendix F). The shared wall pixels occur in both possible gap futures, so a pixelwise mixture of those futures would retain them. This observation rules out that simple pixel-space picture; it does not determine what averaging or distribution shift does in the learned latent space.

A within-encoder counterfactual diagnostic makes the scale of the feature explicit. On 64 pre-reveal contexts for seed 0, the first-step squared distance between wall-present and wall-absent encodings is 0.035, while total teacher-forced prediction error is 0.145. The same ordering holds for all three

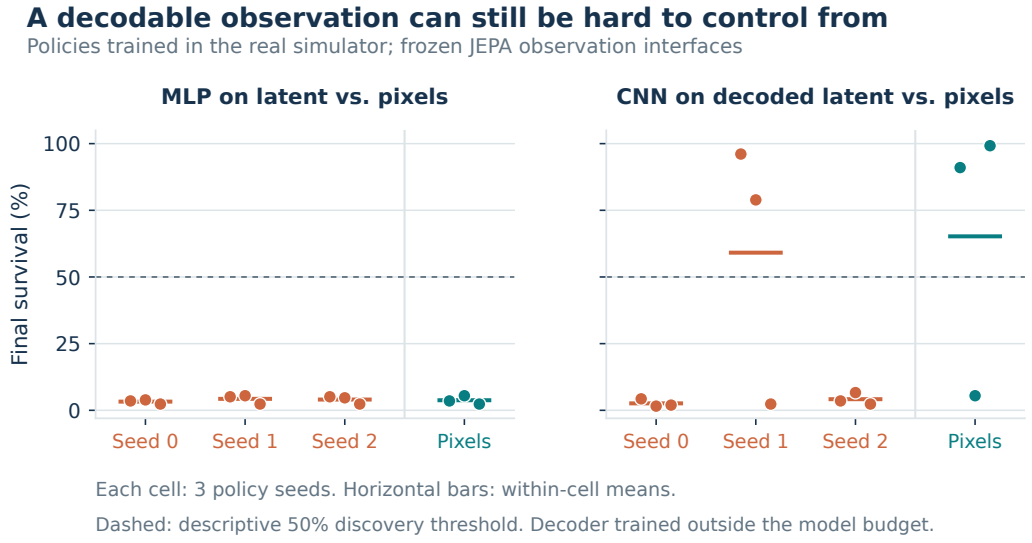


Figure 3: **Recoverable information can support successful control.** Final real-gate survival for MLP policies on frozen latents or pixels (left), and CNN policies on decoded latents or pixels (right). Points are policy runs, grouped by encoder seed; the dashed line is the 50% discovery criterion. Both decoded-latent discoveries use encoder 1. Reconstruction supervision and CNN training are additional to representation fitting.

seeds (Appendix F). This comparison concerns sensitivity to wall removal versus error in all features. It neither identifies a causal mechanism nor permits comparing information content through raw distances across independently fitted latent spaces.

The fitted heads expose another limitation. For encoder 0, termination AUC is 0.96 for border deaths but 0.65 for wall deaths. Predicted rewards for the two death causes are -0.89 and -0.05, respectively, against the true -1. An imagined policy therefore trains on a weak wall-collision signal. The residual-sampling variants retain these heads and the policy protocol. Their poor control outcomes therefore test this particular intervention, not distributional prediction in general.

5 Remove the Wall: Nuisance and Dynamics Interact

Cube worlds reduce the scene to one moving object. Random backgrounds either change independently each frame or remain fixed for an episode; dynamics are deterministic or include hidden wind. A fixed background is predictable but irrelevant to position. Per-frame noise is unpredictable and also irrelevant. These cases distinguish visual complexity from predictive usefulness.

The strongest replicated comparison holds the 512-dimensional grid architecture, dataset size ($500{,}000$ transitions), and training length ($8{,}500$ updates) fixed across four gray-noise conditions. A supervised nonlinear position read-out is fitted on training encodings and evaluated on independent evaluation frames. It achieves subpixel error whenever either the background is persistent or dynamics are deterministic. Combining per-frame noise with wind raises error to 21.33 ± 0.06 pixels across three model initializations (Table 3). This is a failure to expose position through the tested read-out. Its prediction loss remains high, about 0.73 per coordinate, so it is not evidence of a successfully optimized low-loss nuisance solution.

Other fitted models fail at different interfaces. On a black background without wind, the class-token JEPA permits 0.087-pixel position read-out from real encodings, but that read-out has 45.27-pixel error on eight-step predictions. A flow model also fails on persistent color texture, with 18.56-pixel real-latent error. These are single-fit observations with different dataset sizes, reported separately in Appendix B; they preclude a uniform ranking of model families. Together, the cube results motivate

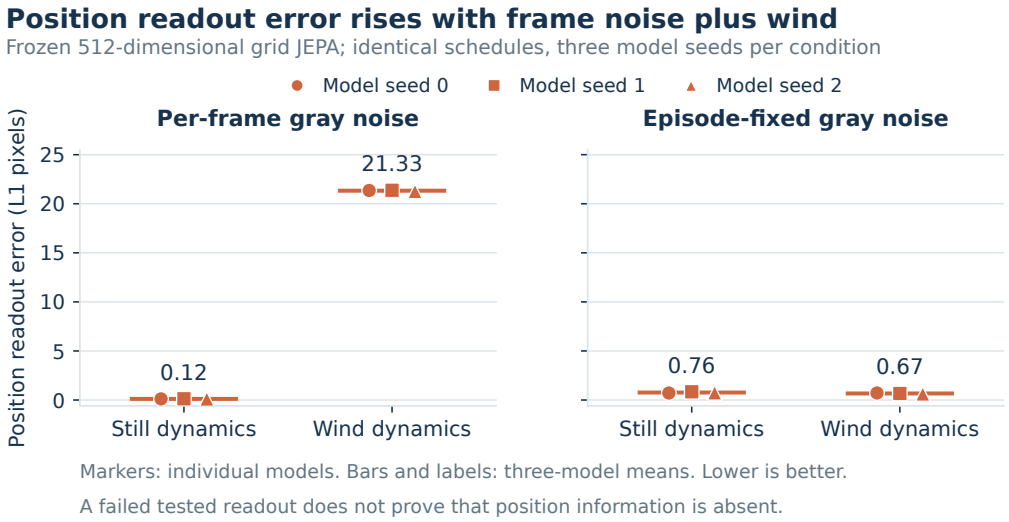


Figure 4: **The matched gray-noise cube comparison.** Position read-out from real encodings for three independently initialized grid JEPAs in each condition. Architecture, latent dimension, update count, and training dataset size are fixed. A high probe error is a failure of the tested read-out, not proof of absent information.

Table 3: **A matched nuisance–dynamics interaction.** Grid-JEPA position L1 error in pixels, mean $\pm$ sample standard deviation across three model fits. The same supervised read-out is applied to real encodings and eight-step predictions; the latter can also reflect read-out extrapolation. Predicted-window evaluation excludes resets and uses overlapping windows.

Gray background / dynamics	Real encoding	Eight-step prediction
Per-frame / deterministic	0.115 ± 0.012	4.84 ± 1.13
Per-frame / wind	21.330 ± 0.057	20.39 ± 0.92
Per-episode / deterministic	0.764 ± 0.068	1.83 ± 0.22
Per-episode / wind	0.668 ± 0.049	10.95 ± 0.53

an objective-level question rather than a universal empirical mechanism: what does low prediction loss plus non-collapse actually guarantee?

6 What the objective does not guarantee

The experiments concern fitted finite networks. The following population statements ask a narrower question: which guarantees follow from prediction error and a Gaussian latent marginal alone? They do not characterize optimization or identify the cause of the empirical failures.

6.1 Non-collapse does not select the controlled state

Write observations as $x_t = R(p_t, b)$, where p_t is controlled state and b is an episode-constant nuisance recoverable from the image. Assume that b is independent of states and actions in the training data. At deployment, assume it is independent of the initial state and transition randomness, and does not directly affect dynamics or reward. Let SIGReg_∞ be the nonnegative population characteristic-function discrepancy, which vanishes for $\mathcal{N}(0, I_D)$ (Appendix A).

Proposition 1 (Gaussian nuisance optimum; extending Sobal et al. (2022))**.** *If the encoder class contains a map $f(R(p, b)) = \varphi(b)$ with $\varphi(b) \sim \mathcal{N}(0, I_D)$ and the predictor class contains the last-latent copy map, their population prediction loss and SIGReg_∞ are both zero. This is a global optimum with covariance I_D and $I(z_t; p_{1:T}) = 0$ under the training distribution. A deployed controller*

observing only these latents, time, and its own actions has the value of a randomized state-blind policy.

The argument is simple: nuisance changes across episodes, satisfying the marginal constraint, but is constant within each episode, making prediction exact. Its deployment value follows by giving a state-blind policy the same private nuisance draw and controller randomness. The Gaussianization assumption requires a rich nuisance and a realizable measurable map; a finite texture does not literally instantiate it. The finite-batch regularizer also differs from the population functional. A complementary innovation bound in Appendix A shows that an encoder preserving all state distances at a fixed scale must incur prediction error for unpredictable state innovations.

6.2 Decodability does not set a useful prediction scale

Even preserving all state information leaves its geometry unconstrained. For any $\varepsilon > 0$, we can place the two values of a relevant bit less than 2ε apart while keeping the *entire* latent distribution exactly standard Gaussian.

Proposition 2 (Exact information at arbitrarily small scale)**.** *For every $\varepsilon > 0$ and $D \geq 1$, there is a measurable observation process with episode-constant nuisance b and a fresh independent fair state bit p_t, and an encoder f_ε, such that $z_t \sim \mathcal{N}(0, I_D)$ and both p_t and b are exactly recoverable almost surely. For fixed b, $0 < \|f_\varepsilon(R(1,b)) - f_\varepsilon(R(0,b))\| < 2\varepsilon$. The optimal next-latent predictor is independent of the next bit and has coordinate-mean squared error in $(0, \varepsilon^2/D]$. With reward $\mathbf{1}\{a_t = p_t\}$, latent feedback attains value 1 per step, while every state-blind policy attains $1/2$.*

The construction partitions one Gaussian coordinate into narrow intervals and uses the bit to select the lower or upper half of each interval's conditional probability mass. The nuisance fixes the interval and the within-half quantile. This preserves the Gaussian marginal and exact decoding while making the two states arbitrarily close. The full proof is in Appendix A. It uses infinite-precision measurable coding; the next bit is intrinsically unpredictable. It therefore separates information from scale, without proving avoidable prediction error or explaining deletion of the predictable gate wall.

Why scale matters for a downstream read-out. For a fixed L-Lipschitz value read-out V and any coupled real and predicted next latents,

$$\left|\mathbb{E}V(Z') - \mathbb{E}V(\widehat{Z}')\right| \leq L\sqrt{\mathbb{E}\|Z' - \widehat{Z}'\|^2}. \tag{2}$$

This standard bound follows from Lipschitz continuity and Cauchy–Schwarz. Prediction error controls value error only relative to the read-out's sensitivity. Exact decodability alone supplies no bound on that sensitivity. We do not estimate L or claim a control guarantee for the fitted networks; the inequality makes explicit the missing bridge between a geometric loss and a decision.

7 Related work

Which JEPA setting? Image and video JEPAs learn by predicting representations (Assran et al., 2023; Bardes et al., 2024). DINO-WM fits dynamics on frozen visual features, and V-JEPA 2 freezes its encoder for action-conditioned prediction with an L1 objective (Zhou et al., 2024; Assran et al., 2025). PLDM instead jointly learns its encoder and dynamics with multiple regularizers (Sobal et al., 2025). Our direct setting is LeWorldModel's jointly trained squared-loss predictor with SIGReg (Maes et al., 2026; Balestriero & LeCun, 2025); the population statements do not automatically transfer to these other objectives.

Predictability, geometry, and planning. Predictable nuisance is an established limitation of joint embeddings (Sobal et al., 2022). Recent work studies how contrastive sampling changes slow-noise bias (Gulati & Nemenman, 2026), and how temporal-distance objectives improve planning-relevant geometry (Bai & Xiong, 2026). LeWorldModel already reports accurate probes alongside weaker planning, while Terver et al. (2026) examine architecture and planner choices and discuss multimodal conditional means. Self-predictive abstraction analyses similarly distinguish predictive structure from task sufficiency (Tang et al., 2023; Ni et al., 2024). Our contribution is a controlled set of representation, rollout, and policy diagnostics with explicit replication, together with a narrow separation between Gaussianity, exact decodability, and loss scale.

Models for decisions. Prediction–control mismatch predates JEPAs (Lambert et al., 2020). Value-aware model learning, value equivalence, and proper value equivalence ask which errors affect downstream Bellman computations and policy values (Farahmand et al., 2017; Grimm et al., 2020; 2021). Expectation models can suffice for suitable value-function classes (Wan et al., 2019); conditional-mean prediction is therefore not inherently incompatible with control. Stochastic imagination models provide another route (Hafner et al., 2020; Alonso et al., 2024; Hafner et al., 2025), but they too can fail at encoding, action conditioning, or rollout stages (Hansen & Wang, 2026). The same interface-level scrutiny should apply to both families.

8 Discussion

The experiments separate capabilities that are often presented together. The Craftax checkpoint supports nearby visual-goal planning, while its reward-planning pipeline has weak held-out reward and value predictions. The gate encoders retain useful obstacle information under an added decoder and CNN, while their tested imagination pipelines fail to discover the strategy. Nuisance cubes and the population constructions show why a marginal non-collapse condition alone cannot settle either question.

The practical consequence is to evaluate a world model at each interface: information recoverable from real encodings, prediction of controlled counterfactuals, head generalization, and behavior in the real environment. A controller's failure should lead to this decomposition before it is used to judge the representation; a successful probe should be followed by prediction and control tests before it is used to certify imagination.

Our empirical scope is one jointly trained squared-loss objective, small visual domains, and short training budgets. Craftax and the flow reference each use one fitted model. Gate predictors, heads, and PPO protocols have not been isolated by a fully matched intervention. The theory gives population existence results, with measurable constructions rather than a finite-network or optimization analysis. In particular, the fresh-bit example does not explain loss of a predictable wall. What the evidence does establish is that non-collapse, decodability, and useful imagined control are separate properties, requiring separate support.

Reproducibility statement

Appendices C–H specify model fitting, post-hoc supervision, evaluation protocols, and replication levels. The tables retain final-checkpoint outcomes rather than selecting intermediate peaks. The accompanying evidence manifest identifies the source artifacts used for each result; independent policy seeds are distinguished from independent model fits. Claims based on a single checkpoint are labeled throughout.

Statement on the use of AI

Language models assisted with prose, code, plotting, proof checking, literature review, and adversarial manuscript review. The authors are responsible for all claims, experiments, and verification of the resulting text and artifacts.

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A PROOFS

A.1 EMPIRICAL OBJECTIVE AND POPULATION IDEALIZATION

For a training batch of B windows and H predicted time positions, the implemented loss is

$$\mathcal{L}_{\text{pred}} = \frac{1}{BHD} \sum_{b,h} \|\widehat{z}_{b,h+1} - z_{b,h+1}\|^2. \tag{3}$$

The notation in Equation 1 averages over time and uses this coordinate-mean convention. One-step targets are teacher-forced at every temporal position, with a shorter history near the start of a window. For encoded windows of T frames, P random unit projection directions u_p, K knots $\tau_k \in [0, 3]$, and $w_k = e^{-\tau_k^2/2}$, the regularizer is

$$R = \frac{B}{TP} \sum_{t,p,k} q_k w_k \left[\left(\frac{1}{B} \sum_b \cos(\tau_k u_p^\top z_{b,t}) - w_k \right)^2 + \left(\frac{1}{B} \sum_b \sin(\tau_k u_p^\top z_{b,t}) \right)^2 \right], \tag{4}$$

where $\Delta = 3/(K-1)$, with $q_k = \Delta$ at the endpoints and 2Δ in the interior. The characteristic function is averaged across batch examples separately at each temporal position; the factor B is part of the implementation with $\lambda = 0.09$.

The population statements use the nonnegative discrepancy

$$\text{SIGReg}_\infty(\mathcal{L}(z)) = \int_{\mathbb{S}^{D-1}} \int_0^3 w(\tau) \left[\big(\mathbb{E}\cos(\tau\langle z, u\rangle) - w(\tau)\big)^2 + \big(\mathbb{E}\sin(\tau\langle z, u\rangle)\big)^2 \right] d\tau \, d\sigma(u), \tag{5}$$

where $w(\tau) = e^{-\tau^2/2}$ and σ is uniform on the sphere. Only the fact that this functional vanishes for a standard Gaussian is needed. It is not a theorem about exact minima of the finite-batch objective.

A.2 NUISANCE OPTIMUM AND THE COST OF INNOVATIONS

Proposition 1.

Proof. Let β recover the nuisance from the observation, set $f^\star = \varphi \circ \beta$, and let $g^\star$ copy the last latent. Then $z_t = \varphi(b) = z_{t+1}$, so every prediction residual vanishes. The Gaussian marginal gives covariance I_D and zero population regularizer. Both objective terms are nonnegative, so their sum is globally minimized at zero. Training-distribution independence gives $I(\varphi(b); p_{1:T}) = 0$.

For deployment, draw nuisance B and the controller's random seed U at reset, independently of the initial state and transition randomness. Run the original controller in the environment with nuisance B. Let a state-blind policy with private seed (B, U) feed constant latent history $\varphi(B)$ to a copy of that controller and reproduce its actions with U. Coupled to the same initial state and transition randomness, the policies take the same first action. If their actions agree through step t, their states, supplied latent histories, and seeds agree, so the next actions agree as well. Induction gives identical trajectories, returns, and terminations. This does not assume independence between B and states subsequently influenced by its induced actions. □

Corollary 1 (Innovation cost for uniformly state-resolving encoders)**.** *Let $p_{t+1} = p_t + \mu(a_t) + w_t$, $\mathcal{H}_t = \sigma(p_{\le t}, b, a_{\le t})$, and $\Sigma_t = \text{Cov}(w_t \mid \mathcal{H}_t)$. Assume finite second moments of states and encoded observations, $x_t = R(p_t, b)$, fixed nuisance b, and $\|f(R(p, b)) - f(R(p', b))\| \ge \kappa \|p - p'\|$ for all admissible p, p', b, with $\kappa > 0$. Every predictor of latent and action history satisfies*

$$\frac{1}{D}\mathbb{E}\|g - f(x_{t+1})\|^2 \ge \frac{\kappa^2}{D}\mathbb{E}\text{tr}\Sigma_t. \tag{6}$$

When $\mathbb{E}\text{tr}\Sigma_t > 0$, this exceeds the nuisance encoder's zero prediction loss under the assumptions of Proposition 1.

The assumption is stronger than injectivity or task sufficiency: every state distance must be preserved at a uniform scale. Innovation is measured after conditioning on state history and nuisance; persistent wind is not automatically fresh innovation.

Corollary 1.

Proof. Put $Z = f(x_{t+1})$ and $\mathcal{C}_t = \sigma(z_{\leq t}, a_{\leq t})$. Because observations and latents are measurable functions of state and nuisance, $\mathcal{C}_t \subseteq \mathcal{H}_t$. Any predictor using a shorter window is also $\mathcal{C}_t$-measurable. Conditional least squares and the tower property give

$$\begin{aligned}\mathbb{E}\|g - Z\|^2 &\geq \mathbb{E}\|Z - \mathbb{E}[Z \mid \mathcal{C}_t]\|^2 \\ &= \mathbb{E}\|Z - \mathbb{E}[Z \mid \mathcal{H}_t]\|^2 + \mathbb{E}\|\mathbb{E}[Z \mid \mathcal{H}_t] - \mathbb{E}[Z \mid \mathcal{C}_t]\|^2 \\ &\geq \mathbb{E}\mathrm{trCov}(Z \mid \mathcal{H}_t).\end{aligned}$$

The cross term vanishes because its first factor has conditional mean zero given $\mathcal{H}_t$. Take conditionally independent copies P, P' of p_{t+1} given $\mathcal{H}_t$. Such copies exist for Euclidean state spaces, using their regular conditional distributions. The nuisance b and current state p_t are fixed under this conditioning. The conditional independent-copy variance identity yields

$$\begin{aligned}\mathrm{trCov}(Z \mid \mathcal{H}_t) &= \tfrac{1}{2}\mathbb{E}[\|f(R(P, b)) - f(R(P', b))\|^2 \mid \mathcal{H}_t] \\ &\geq \tfrac{\kappa^2}{2}\mathbb{E}[\|P - P'\|^2 \mid \mathcal{H}_t] \\ &= \kappa^2 \mathrm{trCov}(p_{t+1} \mid \mathcal{H}_t) = \kappa^2 \mathrm{tr}\Sigma_t.\end{aligned}$$

The final equality follows because $p_t + \mu(a_t)$ is $\mathcal{H}_t$-measurable. Taking expectations and dividing by D proves Equation 6. Averaging these inequalities gives the corresponding bound for a multi-time prediction objective. Proposition 1 supplies the zero-loss comparator when its assumptions also hold. □

A.3 Conditional prediction and geometric scale

Lemma 1 (Conditional mean)**.** *For a fixed encoder and data law with $\mathbb{E}\|z'\|^2 < \infty$, the unique squared-error minimizer over measurable square-integrable predictors from history c and action a is $m(c, a) = \mathbb{E}[z' \mid c, a]$, up to null sets.*

Proof. Let $m = \mathbb{E}[z' \mid c, a]$. For every square-integrable measurable $g(c, a)$, expand $g - z' = (g - m) + (m - z')$. The cross term has expectation zero since $\mathbb{E}[m - z' \mid c, a] = 0$. Hence

$$\mathbb{E}\|g - z'\|^2 = \mathbb{E}\|g - m\|^2 + \mathbb{E}\mathrm{trCov}(z' \mid c, a).$$

The second term does not depend on g. The first is nonnegative and vanishes exactly when $g = m$ almost surely. □

The finite-sample regularizer. The finite-sample regularizer differs from Equation 5. For independent identically distributed samples with characteristic function ϕ and empirical estimate $\widehat{\phi}_n$,

$$\mathbb{E}|\widehat{\phi}_n(t) - w(t)|^2 = |\phi(t) - w(t)|^2 + \frac{1 - |\phi(t)|^2}{n}. \tag{7}$$

Indeed, $\mathbb{E}\widehat{\phi}_n = \phi$ and $\mathbb{E}|\widehat{\phi}_n - \phi|^2 = (1 - |\phi|^2)/n$, since each complex exponential has modulus one; expanding around ϕ gives the identity. At a Gaussian law, the residual equals $(1 - e^{-t^2})/n$. This is estimator bias, not an optimization floor. Changing the law changes both terms; dependent temporal samples introduce additional covariance terms. The implementation multiplies this statistic by batch size B, so its Gaussian finite-batch bias does not shrink as $1/B$ after that scaling. Finite projections and knots also do not characterize Gaussianity.

Proposition 2.

Proof. Partition $\mathbb{R}$ into $I_j = [2\varepsilon j, 2\varepsilon(j + 1))$, $j \in \mathbb{Z}$. Let $q_j = \Pr(G \in I_j)$ for $G \sim \mathcal{N}(0, 1)$. Every q_j is positive and their sum is one. Let F_j be the conditional Gaussian distribution function on I_j, with inverse quantile function Q_j on $(0, 1)$. The conditional CDF is continuous and strictly increasing on the interval interior, so its inverse is measurable and takes values in that interior.

Draw J with probabilities q_j, $U \sim \mathrm{Uniform}(0,1)$, and $V \sim \mathcal{N}(0, I_{D-1})$, independently. Set $b = (J, U, V)$; for $D = 1$, omit V. Take $R(p, b) = (p, b)$ and define

$$f_\varepsilon(R(p,b)) = \big(Q_J((U+p)/2), V\big). \tag{8}$$

The mixture $(U + p_t)/2$ is uniform on $(0, 1)$, since p_t is an independent fair bit. Conditional on $J = j$, the first coordinate therefore has the Gaussian law restricted to I_j. Mixing with weights q_j recovers $\mathcal{N}(0, 1)$. Independence of V gives the joint law $\mathcal{N}(0, I_D)$ and zero population regularizer.

Given the first coordinate y, its interval determines J almost surely. Write $v = F_J(y)$. Then $p_t = \mathbf{1}\{v > 1/2\}$ and $U = 2v - p_t$ almost surely; the other coordinates recover V. Thus the latent recovers both state and nuisance. For fixed b, the two first coordinates are distinct and lie inside the same interval of width 2ε. All remaining coordinates agree, proving the distance bound.

Latent history reveals b, while the next bit remains independent and fair. The next latent consequently has two equiprobable values, say $z^{(0)}(b)$ and $z^{(1)}(b)$. Lemma 1 gives $m(b) = (z^{(0)}(b) + z^{(1)}(b))/2$. Its conditional squared error is $\|z^{(1)}(b) - z^{(0)}(b)\|^2/4$, strictly positive and bounded by ε^2. The midpoint is square-integrable by Jensen's inequality, since both state encodings have finite second moments. Taking expectations and dividing by D proves the coordinate-mean loss bound, including strict positivity since the conditional error is positive almost surely. The midpoint depends only on b, so it is independent of p_{t+1}. Finally, decoding the current bit permits $a_t = p_t$. A state-blind action is independent of the current fair bit, giving success probability $1/2$. □

B THE CUBE WORLDS

The position-readout study crosses black, per-frame random, and per-episode random backgrounds with deterministic or hidden-wind dynamics. Color-noise and black-background datasets contain 100,000 transitions; the gray-noise conditions contain 500,000. Only the four gray grid-JEPA conditions have three independent model initializations. Table 3 and Figure 4 report that replicated comparison. Tables 4 and 5 retain seed-0 outcomes for the broader conditions, using the same canonical benchmark files throughout. Differences across dataset sizes and model recipes are not isolated nuisance effects.

Environment. The scene contains a red 6×6 object in a 63×63 RGB image with a red boundary. Four cardinal actions move the object three pixels. Reset positions are uniformly sampled from integer coordinates in $[16, 42]^2$, and initial wind is zero. In wind conditions, each wind coordinate independently increments by -1, 0, or 1 with probabilities 0.1, 0.8, and 0.1, clipped to $[-2, 2]$; the updated wind is added to the action displacement. Background pixels are uniform integers from 0 to 255; gray noise shares the same sample across RGB channels. Contact with the boundary gives reward -1 and terminates; other rewards are zero, with a 128-step cap.

Read-out fitting. The nonlinear latent position probe is fitted on training data and evaluated on independent `eval.npz` frames. Its spatial softmax-and-offset output is produced by two hidden layers of width 128, trained for 10,000 updates with batch 2,048 and Adam at initial learning rate 10^{-3} with cosine decay. Grid inputs are standardized using training statistics. For class-token inputs, standardization and PCA whitening are selected using the final 10% of fitting rows, then the chosen variant is refitted on the full fitting split. Whitening floors singular values at 1% of the largest. Applying this read-out to predicted latents tests both prediction and read-out extrapolation. The pixel read-out used for decoded flow images is a separate supervised CNN trained with synthetic background replacement using ground-truth cube masks; it is a diagnostic, not a policy or a world-model training signal. Rollout windows are sampled with replacement and overlap, and exclude death or reset during the eight context and 16 forecast frames. The configured benchmark uses 2,048 windows for video metrics and the first 512 for position diagnostics; exact sampled indices were not retained in the archived result files. Surviving windows are a selected subset of wind trajectories, so these errors do not establish death prediction or closed-loop control.

Earlier controller experiments use the three black/color background conditions. Their final survival outcomes are retained below. The class-token JEPA's near-zero loss on per-episode texture and poor reward-driven control are consistent with the known predictable-nuisance mechanism, without exactly realizing Proposition 1. The replicated gray-frame/wind failure instead has high prediction

Table 4: Seed-0 cube position L1 error (pixels) for the grid and class-token JEPAs. "Real" evaluates held-out encoded frames; "Pred. 8" evaluates the same read-out on eight-step predicted latents. Gray conditions use more training transitions than black and color conditions.

Background	Wind	Grid JEPA		Class-token JEPA	
		Real	Pred. 8	Real	Pred. 8
Black	Off	0.097	3.933	0.087	45.268
	On	0.126	11.732	0.049	8.957
Color, frame	Off	0.172	5.570	24.133	19.428
	On	0.253	11.823	25.185	19.317
Color, episode	Off	3.273	6.594	20.804	21.567
	On	1.451	13.342	23.431	21.807
Gray, frame	Off	0.115	4.269	23.954	20.286
	On	21.354	19.696	25.288	19.567
Gray, episode	Off	0.715	1.779	21.316	21.284
	On	0.717	11.553	23.880	23.386

Table 5: Seed-0 flow cube position L1 error (pixels). Latent and pixel columns use different supervised read-outs. The persistent-color failure is visible before dynamics: real encoded position has large error. These are model-recipe references rather than objective-only comparisons.

Background	Wind	Real latent	Pred. 8 latent	Pred. 8 pixels
Black	Off	0.140	0.126	0.242
	On	0.153	11.632	11.622
Color, frame	Off	1.088	1.059	0.395
	On	1.344	12.311	11.985
Color, episode	Off	18.563	21.954	22.423
	On	7.010	18.960	16.937
Gray, frame	Off	3.114	2.667	0.382
	On	3.690	14.392	12.302
Gray, episode	Off	5.062	4.896	1.339
	On	3.066	15.150	13.583

loss and should not be interpreted as the same mechanism. Still-world survival can be achieved by state-blind alternating actions, so it does not certify state feedback.

Table 6: Survival (percent) in the supplementary cube worlds. "LeWM" denotes the class-token JEPA; "grid" denotes the grid JEPA; "goal" and "heads" use latent-goal and learned-return search, respectively. PPO columns use the final saved policies; "px" and "z" identify pixel and latent observations. Still-world survival does not certify state feedback.

Background	Black		Per-frame noise		Per-episode noise	
Dynamics	Still	Wind	Still	Wind	Still	Wind
random	20	0	20	0	20	0
centre	100	99	100	99	100	99
LeWM goal	97	64	5	0	80	0
LeWM heads	100	60	9	0	2	0
LeWM PPO	100	99	0	0	0	0
grid goal	88	1	92	0	100	77
grid heads	100	5	100	0	85	8
grid PPO	100	94	100	0	2	0
flow PPO px	100	94	98	97	98	95
flow PPO z	100	96	99	1	51	1
PPO real	100	98	100	96	100	78

C HYPERPARAMETERS AND COMPUTE

The gate and gray-noise cube conditions supply 500,000 uniform-random transitions each. The earlier black and color-noise cube datasets contain 100,000 transitions. Craftax uses its separate 500,000-transition archive. Model training uses bfloat16 compute, float32 parameters, and one A100. A 500-step calibration estimates the training length fitting 575 seconds including compilation; the grid gate model trains for 8,500 steps (17,000 and 34,000 in the budget sentence of Section 4.2), the class-token cube models for 23,750–24,000. Model seeds 0, 1, 2 of the grid JEPA differ only in the initialisation and data-order seed.

JEPA settings. Both JEPA architectures use batch 128 and AdamW with learning rate and weight decay 10^{-3}, 10% warm-up followed by cosine decay to zero, and gradient clipping at 1.0. SIGReg uses weight 0.09, 17 knots on $[0, 3]$, and 1024 random projections. Neither architecture uses a stop-gradient, teacher, decoder, or reward during world-model fitting. Linear–BatchNorm–GELU–linear projection heads flatten batch, time, and spatial tokens for normalization. Training-mode statistics therefore couple examples and temporal positions; at inference, running statistics define a fixed pointwise projection and the attention predictor is causal. The theoretical causal-map formulation is an inference/population idealization.

	Class-token JEPA	Grid JEPA
encoder	ViT, 7×7 patches	tile transformer + conv
width / depth / heads	192 / 6 / 3	128 / 6 / 8
encoder MLP	768	$4\times$ width
read-out	class token	4×4 grid, 32 channels
latent dimension	192	512
projector	BN projector, width 2048	same, hidden 512
predictor	AdaLN-zero AR	spatial + causal temporal
predictor depth / heads / dim	6 / 16 / 64	6 / 8 / 32
predictor MLP / width	2048 / latent	$2\times$ width / 256
history	3 frames	8 frames
dropout	0.1	0
parameters	15.3M	2.6M + 8.3M

Flow settings. The tokenizer uses 9,000 steps, batch 256, and learning rate 9×10^{-4}; dynamics 16,000 steps, batch 64, the same learning rate, width 256, six layers, eight spatial and temporal attention heads, a 32-frame context, independent per-frame noise levels, a shortcut loss of weight 0.25 every fourth step after the first quarter of training, history dropout 0.25, two history frames resampled every eight steps after half of training, and a temporal loss weight of 0.25. The symlog reward head has 255 bins over $[-20, 20]$; reward and termination losses have weight 0.1 and enter the trunk at full strength for half of training, then at factor 0.1. Inference uses four function evaluations per frame. We corrected terminal-transition masking in the recipe's reward loss (Appendix G).

JEPA read-outs. Gate post-hoc heads use the frozen encoder and the first 800 time rows (80,000 frames) of the same archive used for world-model training. The following 200 time rows are held out from head fitting only; the independent world-model evaluation archive is `eval.npz`. The pixel decoder uses 8,000 steps, batch 256, learning rate 10^{-3}, width 256, six residual blocks; it supports the figures and the decoded-latent control, in which its frozen output, clipped to $[0, 255]$, replaces the pixels of the real-gate PPO; no true pixel enters that policy after the encoder. Reward, termination, and behaviour-value heads are two-layer perceptrons of width 512 on the latent and the one-hot action, 6,000 steps, batch 1024, learning rate 10^{-3}, Monte Carlo value targets at discount 0.99, value loss weight 0.1. The residual predictor models $z_{t+1} - m$ with $m = g(z_{t-7:t}, a_{t-7:t})$ the original grid prediction, conditioned on m, z_t, and the action: three residual blocks of width 1024, 8,000 steps, and batch 2048. The final 20,000 valid windows of the same world-model training archive are excluded from residual fitting and used for its quality check. They are not independent world-model evaluation data. For encoder seed 0, coordinate-mean MSE is 3.07×10^{-4} for the original predictor, 2.05×10^{-4} for one residual sample, and 1.32×10^{-4} for the mean of eight. This

quality check uses eight function evaluations per sample; the reported PPO runs instead use four. Averaging eight learned samples is a random Monte Carlo average, not an exact conditional mean.

Policy optimisation. PPO uses discount 0.99, GAE parameter 0.9, clipping 0.2, value coefficient 0.5, entropy coefficient 0.01, AdamW at 2.5×10^{-4} with weight decay 0.01, four epochs per batch, and gradient clipping at 1.0. Pixels pass through a three-block Impala torso (Espeholt et al., 2018), latents through a two-layer perceptron; a GRU receives that embedding and the previous action; the critic uses 64 categorical bins with histogram-Gaussian targets (Farebrother et al., 2024). Real-environment PPO uses 128 environments and 64 steps per iteration with four minibatch groups, truncated backpropagation over 16 steps, and 1.5M environment steps. In-model PPO uses 150 updates of 512 trajectories of 16 steps (1.2M imagined steps) with minibatch size 128; half of the starts are resets, half dataset states with their real eight-frame context. All reported policy rows use final checkpoints.

Compute. On a shared A100, the seed-0 grid-gate JEPA completes 8,500 updates in 573 seconds and has a logged total of 586.5 seconds. Probe and head fitting takes approximately three minutes. The residual sampler's optimization stage takes 74.3 seconds; model loading, latent extraction, fitting, and quality evaluation together take 275.2 seconds before the final file write. Other logged stages are PPO inside the JEPA 2.0–4.3 min per run; PPO inside the flow model 5–6 min per run at four function evaluations, 20–26 min with the mean of eight; PPO in the real environment 3.7–7.7 min on pixels with the convolutional torso and 51–70 s with the MLP torso, 76–78 s on the frozen latent, 4.7–4.8 min on the decoded latent with the convolutional torso and 1.8–2.9 min with the MLP torso; the latent diagnostic: 2 min.

D Evaluation protocol

Every gate-policy endpoint uses the final checkpoint and 256 real episodes from reset, with the 128-step cap. Survival is the fraction reaching the cap. Walls passed counts walls leaving the frame while the agent remains alive. Discovery is final survival strictly above 50%. Real-training and imagined-training runs advance their random keys under different iteration schedules; their final evaluations are not paired episode trials. Policy seeds are separate runs conditional on a fitted model; independently fitted encoders provide a different replication level. The flow model has only one checkpoint. Reported means weight policy runs equally; standard deviations describe variation across those runs, not uncertainty across model families.

Gate specification. The gate uses the same image and object sizes and cardinal action displacement. A white vertical wall of width three pixels spawns at $x = 60$ every 22 steps and moves left three pixels per step. Its 16-pixel gap is sampled at spawn, top or bottom with equal probability, and becomes visible when the wall reaches $x \leq 40$. The gap location is hidden state before that reveal. Border or wall contact produces reward -1 and termination; the time cap is 128 steps.

Latent diagnostic. For each encoder, the diagnostic uses 64 real contexts with the wall still solid at $x \in \{42, 45\}$. Reconstructed simulator states generate recorded futures and matched wall-free futures under the recorded actions. The recorded branch survives the diagnostic horizon by construction. At each step, wall amplitude is the squared Euclidean distance between encodings of the wall-present and wall-absent frames. The teacher-forced residual uses real encoded history and is a total squared error, summed over latent coordinates. The wall coordinate of a prediction is its projection onto the wall-present minus wall-absent latent difference, with endpoints 0 and 1. Decoded wall intensity averages the minimum colour channel on shared wall pixels, excluding either possible gap band. These measurements compare features within one encoder; raw distances across models are not an information metric.

E Latent-policy torso controls

The MLP on decoded latents passes 0.92 walls per episode with no discoveries in six runs (Table 7), using encoders 0 and 1. The decoded CNN control instead uses all three encoders, with three

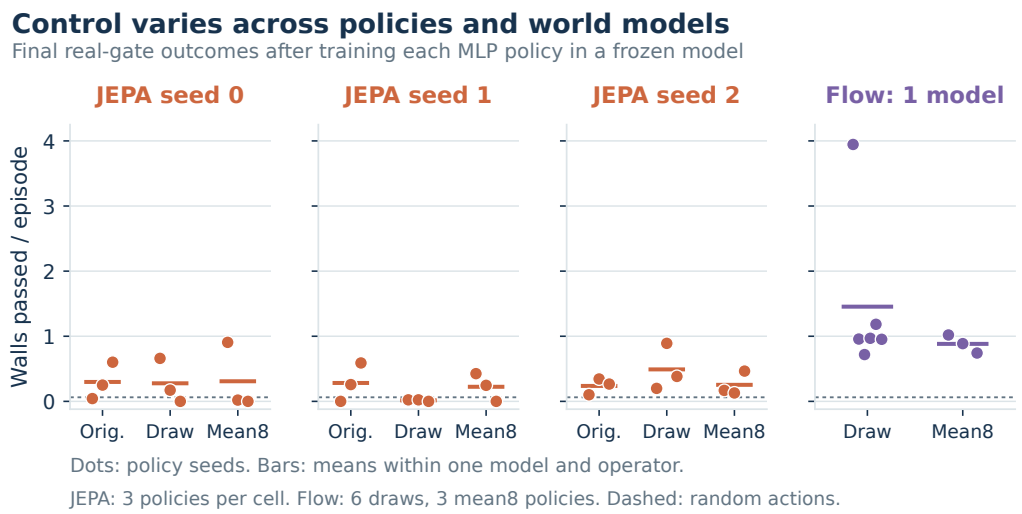


Figure 5: **Control outcomes by model fit and predictor variant.** Each point is a final policy evaluated on 256 real episodes. Original prediction, residual sampling, and residual means are separate conditions; model and policy replication are distinct. Horizontal bars show within-model means, and the dashed line marks random actions. The flow reference uses one checkpoint.

policies per encoder. Both of its discoveries use encoder 1. The decoder is frozen and reconstruction-supervised; success therefore establishes useful encoded information under this added read-out, not sufficiency of the unsupervised training objective. Failure of the true-pixel MLP recipe establishes no upper bound on MLP capacity.

Table 7: Real-gate observation and torso controls: final walls per episode and survival, mean ± standard deviation over policies; discovery means final survival above 50%. All policies use 1,499,136 real transitions.

observation of the real-gate PPO	runs	walls passed	survival (%)	discovered
pixels, convolutional torso	3	2.91±1.65	65±52	2/3
pixels, MLP torso	3	0.91±0.08	4±2	0/3
frozen JEPA latent, MLP torso	9	0.94±0.04	4±1	0/9
decoded JEPA latent, convolutional torso	9	1.57±1.26	22±37	2/9
decoded JEPA latent, MLP torso	6	0.92±0.08	3±2	0/6

F CONTROL OUTCOMES AND LATENT DIAGNOSTICS

Table 8 gives aggregate final outcomes; Table 9 lists every imagined JEPA policy. The pooled imagined JEPA mean weights all original, sampled-residual and mean-residual runs equally. The flow rows distinguish six sampled policies from three policies using a mean of eight, all sharing the same model fit. They do not estimate variation across independent flow-model fits. Tables 10 and 11 retain the encoder and geometry diagnostics supporting the gate observations.

Table 8: Final control outcomes in the real gate, mean ± standard deviation over policy runs. Imagined training uses 1,228,800 transitions per policy; real training uses 1,499,136. Model-fit counts and policy-run counts identify distinct replication levels.

Controller	Models	Policies	Walls	Survival (%)	Discovery
JEPA: original	3	9	0.27 ± 0.22	0 ± 1	0/9
JEPA: mean of 8	3	9	0.26 ± 0.30	1 ± 1	0/9
JEPA: one sample	3	9	0.26 ± 0.32	1 ± 1	0/9
Flow: mean of 8	1	3	0.88 ± 0.14	3 ± 1	0/3
Flow: one sample	1	6	1.45 ± 1.23	19 ± 39	1/6
Random	–	–	0.06	0	–
Alternating	–	–	0.02	0	–
Real pixels: CNN	–	3	2.91 ± 1.65	65 ± 52	2/3
Real pixels: MLP	–	3	0.91 ± 0.08	4 ± 2	0/3
Real JEPA: MLP	3	9	0.94 ± 0.04	4 ± 1	0/9
Real decoded: CNN	3	9	1.57 ± 1.26	22 ± 37	2/9

Table 9: Every imagined JEPA policy at its final checkpoint: survival (percent) and walls per episode by encoder seed, predictor variant and PPO seed. Each cell uses 256 real episodes.

		Survival (%)			Walls passed		
Predictor	Model	PPO 0	PPO 1	PPO 2	PPO 0	PPO 1	PPO 2
Original	0	0	0	1	0.04	0.25	0.60
	1	0	0	1	0.00	0.26	0.59
	2	0	1	0	0.10	0.34	0.27
Residual mean	0	3	0	0	0.91	0.02	0.00
	1	2	1	0	0.43	0.25	0.00
	2	0	0	1	0.17	0.13	0.46
Residual sample	0	2	1	0	0.66	0.17	0.00
	1	0	0	0	0.02	0.02	0.00
	2	0	3	1	0.20	0.89	0.38

Table 10: Grid-JEPA diagnostics by encoder seed. Wall amplitude and total teacher-forced residual are squared distances at future steps 1 and 8; decoded intensity uses steps 1 and 2. The wall coordinate has endpoints 0 (wall absent) and 1 (true encoding). Head AUC compares each death cause with alive transitions; PPO wall counts pool predictor variants.

JEPA encoder seed	0	1	2
Wall amplitude, step 1	0.035	0.028	0.088
Wall amplitude, step 8	0.124	0.082	0.193
Total residual, step 1	0.145	0.130	0.156
Total residual, step 8	0.249	0.228	0.279
Wall coordinate, step 1	0.39	0.45	0.66
Decoded wall, step 1	0.00	0.00	0.68
Decoded wall, step 2	0.00	0.00	0.02
Border-death AUC	0.96	0.96	0.88
Wall-death AUC	0.65	0.65	0.76
Imagined PPO, walls	0.29±0.34	0.17±0.22	0.33±0.24

Table 11: JEPA geometry and decoded wall intensity for model seed 0, averaged over 64 pre-reveal contexts. Distances use matched actions and the same encoder; total residuals include all latent coordinates. Wall intensity is measured on shared wall pixels.

Future step	1	3	5	8
Squared distance, summed over coordinates				
Wall-present versus wall-absent	0.035	0.041	0.071	0.124
Total teacher-forced residual	0.145	0.152	0.200	0.249
Residual along wall direction	0.015	0.021	0.048	0.104
Residual orthogonal to wall direction	0.130	0.132	0.152	0.145
Predicted wall coordinate	0.39	0.33	0.29	0.21
Decoded wall intensity				
Encoded true future	1.00	1.00	1.00	1.00
Original JEPA rollout	0.00	0.00	0.00	0.00

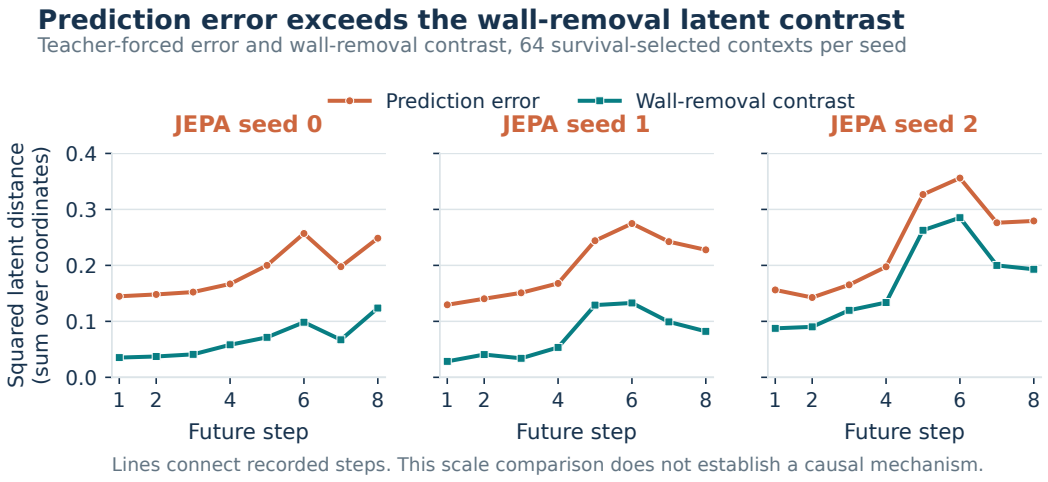


Figure 6: **Within-encoder scale diagnostics.** For each of the three gate encoders, wall-removal distance and teacher-forced total one-step residual are summed squared latent distances. Curves average 64 pre-reveal contexts whose recorded futures survive the diagnostic horizon. These are feature sensitivity and total prediction error, not commensurate measures of information content.

G TERMINAL REWARDS IN THE FLOW RECIPE

The original recipe masks transitions entering reset frames out of reward training. Here the only nonzero reward occurs on terminal transitions, so that mask removes the reward signal. The corrected head predicts the next reward from the current frame summary and action with terminal transitions retained in its loss. The reported flow-policy outcomes use this corrected convention. The single-checkpoint reference therefore denotes the complete tokenizer, dynamics, reward and termination package, rather than an isolated change in transition stochasticity.

H CRAFTAX PROTOCOLS AND SINGLE-CHECKPOINT RESULTS

The grid checkpoint uses training seed 0, a tile encoder, $16 \times 32 = 512$ latent coordinates, a six-layer factored predictor of width 256, history length eight, 17 actions, batch 128, and 8,500 updates. The recorded parameter count is 11.12 million. Training takes 604.5 seconds from loading through the logged end-to-end timer (578.5 seconds in the update loop); thus it is an approximately ten-minute run, not a strict sub-600-second result. The frozen decoder and heads add 175.6 seconds. Results are from one model fit and evaluation seed 0.

Goal generation and planning. Reset states receive seven random warm-up actions. Six further random actions generate a reachable goal image. CEM optimizes a length-six action sequence in latent space, with population 512, 32 elites, and six iterations. The learned planner uses the goal image, without privileged position coordinates or the generating actions in its cost. It executes the entire selected sequence without observation feedback. Success is equality of final player positions. Goal generation, random actions, and the selected plan use the same future simulator random keys; replaying the generating actions therefore reaches all goals and serves only as a protocol check. This protocol samples nearby reachable goals, not arbitrary distant tasks.

The planned endpoint position L1 error is 0.0781, against 1.5313 for random actions. Pixel MSE on the 0–255 scale is 104.00 versus 1477.92. These quantities use the same 128 starts as the reported $121/128$ versus $24/128$ reach counts. They do not quantify variation across fitted models.

Reward control and head generalization. Reward and value heads use the first 80,000 transitions (320 time rows across 250 streams) from the world-model training archive. Held-out time rows are excluded from head fitting but not from encoder training. Value fitting uses the first 80 time rows owing to truncated-return handling. Reward and value fitting correlations are 0.922 and 0.995; held-out correlations are 0.353 and 0.266. Held-out reward RMSE is 0.222. Mean predicted held-out value is 2.356, versus mean empirical discounted return 1.348.

Reward evaluation uses 128 starts, 100 steps, no autoreset, and rewards accumulated while alive. The CEM controller executes one action and replans from each real observation. Random actions, myopic fitted-reward argmax, one predicted step plus value, and CEM obtain returns 1.4453, 0.7813, 0.9438, and 0.4734, respectively; their across-start standard deviations are 1.0897, 1.1109, 1.2014, and 1.1039. Fractions alive at the end are 0.6328, 0.7266, 0.6328, and 0.6719. Aggregate summaries do not preserve the per-start covariance needed for paired confidence intervals. The archive's mean reward multiplied by 100 is not an online behavior baseline and is excluded from the comparison.

Earlier Craftax experiments used a different 81×32 spatial latent and different horizons. They are not merged with these 512-dimensional results. There is no matched online flow-controller comparison for this checkpoint.